t3-w09 — condition C3

Need for statistic — Need for statistics: purposes, and the collection and tabulation of data

> C2 plus a validation and repair pass over the draft.

INPUT LAYERS

Input layer	What it carries	Channel	In this condition
Scope	what is to be taught, and to whom	prompt	**present**
Prior knowledge	what the learner already knew coming in	prompt	**present**
Materials	what the lesson was represented with	prompt	**present**
Position	where this objective sat in the sequence	prompt	**present**
Method	the shape of the teaching, as event labels	prompt	**present**
Enacted content	what was actually explained and worked through	prompt	**present**
Set work	what the learners were asked to do in class	prompt	absent
Assessment	the questions the lesson was checked with	prompt	**present**
Figures	the drawings the lesson used	tool	absent

WHAT THE MODEL IS SENT

Contract `ComposeWriteGroundedSection`, declaring: `artefact_spec, lesson_context, unit_context, lesson_content`.

Lesson context — shared by every section

`subject`

```
Mathematics
```

`grade_level`

```
JSS1
```

`topic`

```
Need for statistic
```

`subtopic`

```
Need for statistics: purposes, and the collection and tabulation of data
```

`term`

```
Third Term
```

`week`

```
9
```

`lesson_objectives`

• explain the purpose of statistic
• analyse and interpret statistical data presented in tables

entry_behaviour

• Students are familiar with counting and recording information.

instructional_materials

• classroom register, bottle tops, pencils or counters.

Objective 1 — explain the purpose of statistic

objective_text

explain the purpose of statistic

current_step

1

instructional_events

• present_content

assessment_points

• (a) Define statistic

lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "statistics is the science of collecting, organizing, presenting, analyzing and
interpreting\ndata.\n\n#### Meaning of Data\n\nData are facts or information collected for a
particular purpose.\n\n#### Importance of statistics\n\nIt helps us to\n\n- (a) Collect
Information\n- (b) organize Information\n- (c) Compare Information\n- (d) Make decisions\n- (e)
Present Information Clearly."
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 2 — analyse and interpret statistical data presented in tables

`objective_text`

analyse and interpret statistical data presented in tables

`current_step`

2

`assessment_points`

• (b) State three methods of collecting data.
• (c) Prepare a frequency table for the following data: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.

`lesson_content`


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Collection of Data\n\nData can be collected through\n\n1. observation\n2.
counting\n3. Interview\n4. Questionnaire.\n\n#### Tally Marks\n\nTally Marks are used to record
data as they are collected.\n\n**Example:** The fig below show the same data collected in two
different ways by two student\n\n(a) bus, car, car, car, lorry, bicycle, bicycle, car, car,
lorry, bicycle, car.\n\nThe above data can be represented in Tally\n\n| Vehicles | Tally | Total
|\n|---|---|---|\n| car | \\|\\|\\|\\|/ \\| | 6 |\n| bus | \\| | 1 |\n| lorry | \\|\\| | 2 |\n|
bicycle | \\|\\|\\| | 3 |\n| Motor Bike | | |\n\n#### Frequency Table\n\nA frequency table is a
table showing how many times each item occurs in a set of data.\n\n**Examples:**\n\nFavourite
colours of student:\n\n| colour | frequency |\n|---|---|\n| Red | 18 |\n| Blue | 20 |\n| Green |
14 |\n| Yellow | 16 |\n| White | 15 |\n| Pink | 30 |"
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `instructional_events`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

WHAT THE MODEL PRODUCED

Objective 1

Why We Use Statistics

> EXPLANATION
> Statistics is the science of collecting, organising, presenting, analysing and interpreting data.
>
> **Data** are facts or pieces of information gathered for a particular purpose.
>
> **Why statistics matters**
>
> - **Collect information** – it helps us gather the facts we need.
> - **Organise information** – it puts the data in a clear order so we can see patterns.
> - **Compare information** – it lets us see how different sets of data relate to each other.
> - **Make decisions** – it gives a solid basis for choosing what to do.
> - **Present information clearly** – it turns numbers into tables, charts or summaries that are easy to understand.

> WORKED_EXAMPLE
> **Worked example**
>
> Imagine you are the class monitor and you want to know how many students prefer chocolate, vanilla or strawberry ice-cream. You ask each student and write down the answers.
>
> 1. **Collect** the answers – you write down each student's favourite flavour.
> 2. **Organise** the data – you count how many chose each flavour.
> 3. **Compare** – you see which flavour is most popular.
> 4. **Decide** – you can decide which flavour to buy for the class party.
> 5. **Present** – you make a simple table showing the numbers for each flavour, so everyone can see the result clearly.

1. 1. In your own words, write a definition of *statistics*.
answer: Statistics is the science of collecting, organising, presenting, analysing and interpreting data.

2. 2. List the five main purposes of statistics mentioned above.
answer: 1) Collect information, 2) Organise information, 3) Compare information, 4) Make decisions, 5) Present information clearly.

3. 3. Your school wants to know the most popular sport among students. Explain how statistics can help them achieve this.
answer: They can ask each student to name their favourite sport (collect), count how many chose each sport (organise), see which sport has the highest count (compare), decide to support that sport (make decisions), and show the results in a table or chart (present).

Objective 2

Understanding Data Collection and Making Frequency Tables

EXPLANATION

Statistics helps us make sense of information by organising, summarising and interpreting data. Before we can analyse data, we need to collect it in a systematic way. Common ways of gathering data are:

1. **Observation** – watching and noting what happens (e.g., counting the number of different vehicles that pass a road).
2. **Counting** – simply tallying items that can be enumerated (e.g., how many students have each favourite colour).
3. **Interview** – asking people questions and recording their answers.
4. **Questionnaire** – giving a set of written questions for people to complete.

When data are collected, we often use **tally marks** to record each occurrence as it happens. A group of five tally marks is written as four vertical lines \|\|\|\| and a diagonal slash \|/ to make it easy to count quickly.

Once the data are gathered, we can organise them in a **frequency table**. A frequency table lists each distinct item (or class) and shows how many times it occurs in the data set. This makes it simple to see which items are most common and to compare quantities.

Below is an example of a frequency table that shows the number of each type of vehicle observed:

Vehicle	Tally	Total
Car	|||||/ |	6
Bus	|	1
Lorry	||	2
Bicycle	|||	3
Motor Bike		0

The **Total** column tells us how many times each vehicle was seen.

WORKED_EXAMPLE

Worked Example: Creating a Frequency Table

Data collected: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2

1. List the different numbers that appear: 1, 2, 3, 4.
2. Count how many times each number occurs:
   - 1 appears 3 times
   - 2 appears 3 times
   - 3 appears 4 times
   - 4 appears 3 times
3. Write the frequency table:

Value	Frequency
1	3
2	3
3	4
4	3

This table shows the distribution of the data and can be used for further analysis.

1. (b) List three different methods you could use to collect data about the favourite sports of students in your class.
answer: Observation, questionnaire, interview

2. (c) Prepare a frequency table for the following data set: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.
answer: Value | Frequency 1 | 3 2 | 3 3 | 4 4 | 3

LAYER BY LAYER: WHAT WAS GROUNDED AND WHAT WAS NOT

Output layer	Made	Written from	Verdict
Title the reader-facing framing of the section	2	Scope	**grounded**
Explanation the principle, rule or relationship to review	2	Enacted content	**grounded**
Worked example a problem with its solution worked through	2	Enacted content	**grounded**
Figure a drawing the learner reads from	0	nothing in the input	not produced
Practice what the learner does	5	Assessment written from the teacher's evaluation questions instead of her set work	**substituted**
Answer what the learner checks against	5	Enacted content	**grounded**